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Inventor(s)	Sandberg; Stuart D. et al.

Avoiding contention between synchronization packets and in-phase and quadrature (IQ) packets communicated over fronthaul of a radio access network

Abstract

Techniques are described for avoiding contention between synchronization packets (for example, Institute of Electrical and Electronics Engineers (IEEE) 1588 Precision Time Protocol (PTP) synchronization packets) and in-phase and quadrature (IQ) packets communicated over a fronthaul network used in a radio access network (for example, a Fifth Generation (5G) radio access network).

Inventors: Sandberg; Stuart D. (Acton, MA), Salahuddeen; Irfaan Ahamed (Acton, MA)

Applicant: Outdoor Wireless Networks LLC (Richardson, TX)

Family ID: 75974794

Assignee: Outdoor Wireless Networks LLC (Richardson, TX)

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Primary Examiner: Sefcheck; Gregory B

Attorney, Agent or Firm: Fogg & Powers LLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Patent Application Ser. No. 62/940,767, filed on Nov. 26, 2019, which is hereby incorporated herein by reference in its entirety.

BACKGROUND

(1) The 3GPP fifth generation (5G) radio access network (RAN) architecture includes a set of base stations (also referred to as “gNBs”) connected to the 5G core network (5GC) and to each other. Each gNB comprises three entities—a centralized unit (CU), a distributed unit (DU), and a set of one or more radio units (RUs). The CU can be further split into a CU user plane entity (CU-UP) and a CU control plane entity (CU-CP). The functions of the RAN can be split among these entities in various ways. For example, the functional split between the DU and the RUs can be configured so that the DU implements some of the Layer-1 processing functions (for the wireless interface) and each RU implements the Layer-1 functions that are not implemented in the DU as well as the basic RF and antenna functions. The DU is coupled to each RU using a fronthaul network (for example, one implemented using a switched Ethernet network) over which data is communicated between the DU and each RU including, for example, user-plane data (for example, in-phase and quadrature (IQ) data representing time-domain or frequency-domain symbols).

(2) Each RU is synchronized with its associated DU. That is, the local clock in each RU is synchronized with the local clock in the associated DU. One way this can be done is by synchronizing the DU and each RU to the same timing master. This synchronization can be done using the Institute of Electrical and Electronics Engineers (IEEE) 1588 Precision Time Protocol (PTP), where the timing master serves as a PTP grandmaster. When PTP is used, two types of PTP messages or packets are communicated over the fronthaul network. Synchronization (Synch) packets are communicated from the timing master to each DU or RU and Delay Request (DelReq) packets are communicated from each DU or RU to the timing master.

(3) For best synchronization performance, a number of the Synch packets communicated each second to each DU or RU should not suffer contention delay with other traffic on the fronthaul network. Such contending traffic includes IQ data transmitted from the DU to the RUs. The DU typically transmits a burst of IQ packets for each 5G slot on the fronthaul, each burst transmission starting at approximately the same time relative to the start of the respective 5G slot. Without loss of generality in what follows we assume each IQ burst starts at the beginning of its corresponding slot.

(4) The slot rate used in 5G can be specified by the following:

$$R_{\text{slot}} = 1000 * 2^{\text{sup.v}}$$

(5) where R_{slot} is the slot rate (in slots per second), v is the 5G numerology (for example, where a numerology of 3 ($v=3$) for 120 kHz subcarrier spacing). The slot rate has an associated slot period specifying a duration of each slot. A PTP grandmaster typically transmits Synch packets at a rate of predetermined number of packets every second (represented here by the variable N_s). Therefore, there is $1/N_s$ seconds between consecutive Synch packets transmitted by the PTP grandmaster. As a result, the number of slots (represented here by the variable M) between consecutive Synch packets is $M = (1000 * 2^{\text{sup.v}}) / N_s$. In general, the number of slots between consecutive Synch packets M is not an integer.

(6) Specifications promulgated by the Open Radio Access Network (O-RAN) Alliance recommend the use of a Synch packet transmission rate N_s that is a power of 2. That is, the recommended Synch packet transmission rate can be represented as:

$$N_s = 2^{\{\text{circumflex over ()}\}k}$$

(7) where $k \in \{0, 1, 2, \dots, 7\}$. As a result, the number of slots between consecutive Synch packets M can be represented as:

$$M = 1000 * 2^{\text{sup.v}} * (v - k)$$

(8) The n th Synch packet is transmitted into the fronthaul network during the current slot at the

following time (in general a fractional (non-integer) number of slots):

$$Arr_n = \theta + n \cdot M$$

(9) where θ is an arbitrary “phase” difference between timing master and the DU and is $0 \leq \theta < M$. Note that M will be an integer when $k \leq v+3$. Taking a numerology of 3 (that is, $v=3$), for each recommended rate in $\{1, 2, 4, 8, 16, 32, 64\}$, M is an integer and all Synch packets are transmitted into the fronthaul at the same time, modulo the slot duration of 125 microseconds (us). For a Synch packet transmission rate N_s of 128 packets per second, the number of slots between consecutive Synch packets M would equal $1000/16$ or 62.5 slots and Synch packets would be transmitted at only two distinct times within any given slot (that is, either at the beginning of the slot or at the midpoint of the slot, offset by θ). If the transmission time (duration) for the burst of IQ packets for each slot is $T > 0.5$ slots and $0 \leq (\theta \bmod 1) \leq T - 0.5$, it is seen that all of the 128 synchronization packets will experience contention with IQ packets. This will impair the performance of the PTP synchronization process. This will occur because, as stated earlier, the DU will transmit a new batch of IQ packets on the fronthaul for each 5G slot, with those IQ packets being transmitted at approximately the same point in each 5G slot. Note in the above discussion that if each IQ burst were to instead start with a nonzero delay relative to the start of the slot, and if there were different propagation delays for Synch packets than for the IQ burst to a given switch, the above condition $0 \leq (\theta \bmod 1) \leq T - 0.5$ for contention at the switch would still apply, but with a shifted interval, i.e., $0 + \tau \bmod 1 \leq (\theta \bmod 1) \leq T - 0.5 + \tau \bmod 1$, for τ depending on IQ burst start time, and the propagation delays.

SUMMARY

(10) One embodiment is directed to a timing master for use in a radio access network (RAN) comprising a plurality of nodes, at least some of which are communicatively coupled to each other via a fronthaul network. The RAN is configured to wirelessly communicate with user equipment (UE) using a predetermined slot rate having an associated slot period specifying a slot duration of each slot. The timing master comprises one or more network interfaces to communicatively couple the timing master to the fronthaul network, at least one programmable device coupled to the one or more network interfaces, a global navigation satellite system (GNSS) receiver coupled to the at least one programmable processor; and a local clock coupled to the at least one programmable device. The timing master is configured to communicate synchronization packets from the timing master to at least some of the nodes of the RAN over the fronthaul network at a synchronization packet transmission rate, the synchronization packet transmission rate specifying a number of synchronization packets transmitted each second. Each of said number of synchronization packets transmitted each second is transmitted at a respective transmission time offset relative to the slot period. The timing master is configured so that the respective transmission time offsets modulo the slot duration for said number of synchronization packets transmitted during each second are unique and distributed throughout the slot period.

(11) Another embodiment is directed to a system comprising a radio access network (RAN) comprising a plurality of nodes, at least some of which are communicatively coupled to each other via a fronthaul network. The system also comprises a timing master communicatively coupled to the nodes of the RAN via the fronthaul network. The RAN is configured to wirelessly communicate with user equipment (UE) using a predetermined slot rate having an associated slot period specifying a slot duration of each slot. The system is configured to communicate synchronization packets from the timing master to at least some of the nodes of the RAN over the fronthaul network at a synchronization packet transmission rate, the synchronization packet transmission rate specifying a number of synchronization packets transmitted each second. Each of said number of synchronization packets transmitted each second is transmitted at a respective transmission time offset relative to the slot period. The system is configured so that the respective transmission time offsets modulo the slot duration for said number of synchronization packets transmitted during each second are unique and distributed throughout the slot period.

- (12) Another embodiment is directed to a method of operating a radio access network (RAN) comprising a plurality of nodes, at least some of which are communicatively coupled to each other via a fronthaul network, the method comprises wirelessly communicating with user equipment (UE) using a predetermined slot rate having an associated slot period specifying a slot duration of each slot and communicating synchronization packets from the timing master to at least some of the nodes of the RAN over the fronthaul network at a synchronization packet transmission rate, the synchronization packet transmission rate specifying a number of synchronization packets transmitted each second. Each of said number of synchronization packets transmitted each second is transmitted at a respective transmission time offset relative to the slot period. The respective transmission time offsets modulo the slot duration for said number of synchronization packets transmitted during each second are unique and distributed throughout the slot period.
- (13) Another embodiment is directed to a radio access network (RAN) system comprising a plurality of nodes, at least some of which are communicatively coupled to each other via a fronthaul network. The RAN system is configured to wirelessly communicate with user equipment (UE). The system is configured so that, while at least one of the nodes of the RAN system receives synchronization packets transmitted from a timing master over the fronthaul network at a synchronization packet transmission rate and synchronization packet phase, said at least one of the nodes discovers the synchronization packet transmission phase of the synchronization packets being transmitted from the timing master and selectively delays transmission of at least some in-phase and quadrature (IQ) data over the fronthaul network in order to avoid contention with at least some of the synchronization packets transmitted from the timing master over the fronthaul network.
- (14) Another embodiment is directed to a method of operating a radio access network (RAN) comprising a plurality of nodes, at least some of which are communicatively coupled to each other via a fronthaul network. The RAN is configured to wirelessly communicate with user equipment (UE). The method comprises, while at least one of the nodes of the RAN system receives synchronization packets transmitted from a timing master over the fronthaul network at a synchronization packet transmission rate and a synchronization packet transmission phase, discovering, by said at least one of the nodes, the synchronization packet transmission phase of the synchronization packets being transmitted from the timing master and selectively delaying, by said at least one of the nodes, transmission of at least some in-phase and quadrature (IQ) data over the fronthaul network in order to avoid contention with at least some of the synchronization packets transmitted from the timing master over the fronthaul network.
- (15) Other embodiments are disclosed.
- (16) The details of various embodiments are set forth in the accompanying drawings and the description below. Other features and advantages will become apparent from the description, the drawings, and the claims.
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Description

DRAWINGS

- (1) FIG. 1 is a block diagram illustrating one exemplary embodiment of a centralized radio access network (C-RAN).
- (2) FIG. 2 comprises a high-level flowchart illustrating one exemplary embodiment of a method of operating a radio access network (RAN) in which contention between IQ packets and at least some synchronization packets communicated over the fronthaul is avoided.
- (3) FIG. 3 comprises a high-level flowchart illustrating one exemplary embodiment of a method of operating a RAN in which contention between IQ packets and at least some synchronization packets communicated over the fronthaul is avoided.
- (4) Like reference numbers and designations in the various drawings indicate like elements.

DETAILED DESCRIPTION

(5) FIG. 1 is a block diagram illustrating one exemplary embodiment of a centralized radio access network (C-RAN) **100**. The C-RAN **100** is also referred to here as a “C-RAN system” **100**, “RAN system” **100**, “RAN” **100**, or just “system” **100**). The C-RAN **100** shown in FIG. 1 implements a base station. The C-RAN system **100** can also be referred to here as a “base station” or “base station system.”

(6) The example shown in FIG. 1 is described here as being configured to implement a 5G RAN architecture in which the base station **100** comprises, for each cell **102** served by the base station **100**, a central unit (CU) **104**, a distributed unit (DU) **106**, and multiple remote units (RUs) **108**. Each remote unit **108** is remotely located from the CU **104** and, in the example shown in FIG. 1, the DU **106**. Also, in this exemplary embodiment, at least one of the RUs **108** is remotely located from at least one other RU **108**. Each RU **108** includes or is coupled to one or more antennas **110** via which downlink RF signals are radiated to various items of user equipment (UE) **112** and via which uplink RF signals transmitted by UEs **112** are received.

(7) The system **100** is coupled to a core network **114** of the associated wireless network operator over an appropriate back-haul **115** (such as the Internet). Also, each DU **106** is communicatively coupled to the RUs **108** served by it using a front-haul network **116**. The DU **106** and the RUs **108** include one or more network interfaces (not shown) in order to enable the DU **106** and RUs **108** to communicate over the front-haul network **116**.

(8) In one implementation, the front-haul **116** that communicatively couples each DU **106** to the RUs **108** is implemented using a switched ETHERNET network. In such an implementation, each DU **108** and RU **108** includes one or more ETHERNET interfaces for communicating over the switched ETHERNET network used for the front-haul **116**. However, it is to be understood that the front-haul between each DU **106** and the RUs **108** served by it can be implemented in other ways.

(9) Generally, for each cell **102** implemented by the C-RAN **100**, the corresponding CU **104** and DU **106** serving the cell **102** performs the LAYER-3 and LAYER-2 functions for the particular wireless interface used for that cell **102**. Also, for each cell **102** implemented by the C-RAN **100**, the corresponding DU **106** serving the cell **102** performs some of the LAYER-1 functions for the particular wireless interface used for that cell **102**. Each of the RUs **108** serving that cell **102** perform the LAYER-1 functions not performed by the DU **106** as well as implementing the basic RF and antenna functions.

(10) In the particular embodiment shown in FIG. 1, the C-RAN **100** is configured to use one or more Fifth Generation (5G) wireless interfaces, associated protocols, and architectures. However, it is to be understood that other embodiments can be implemented in other ways—for example, the C-RAN **100** can be configured to use other wireless interfaces, protocols, and architectures such as the Long-Term Evolution (LTE) wireless interfaces, protocols, architectures and/or to support multiple wireless interfaces, protocols, and architectures (for example, to support both 5G and LTE wireless interfaces, protocols, and architectures).

(11) In the exemplary embodiment shown in FIG. 1, a management system **118** is communicatively coupled to the CU **104**, DU **106**, and the RUs **108**, for example, via the Internet (or another network used to implement the back-haul **115** to the core network **114**) and an Ethernet network (for example, an Ethernet network used to implement the front-haul network **118**).

(12) Each CU **104**, DU **106**, RU **108**, and management system **118** (and the functionality described as being included therein), as well as the system **100** more generally, and any of the specific features described here as being implemented by any of the foregoing, can be implemented in hardware, software, or combinations of hardware and software, and the various implementations (whether hardware, software, or combinations of hardware and software) can also be referred to generally as “circuitry” or a “circuit” configured to implement at least some of the associated functionality. When implemented in software, such software can be implemented in software or firmware executing on one or more suitable programmable processors or configuring a

programmable device (for example, processors or devices included in or used to implement special-purpose hardware, general-purpose hardware, and/or a virtual platform). Such hardware or software (or portions thereof) can be implemented in other ways (for example, in an application specific integrated circuit (ASIC), etc.). Also, the RF functionality can be implemented using one or more RF integrated circuits (RFICs) and/or discrete components. Each CU **104**, DU **106**, RU **108**, and management system **118** and the system **100** more generally, can be implemented in other ways.

(13) In the embodiment shown in FIG. **1**, a timing master **120** is used to synchronize one or more nodes of the RAN **100**. The timing master **120** can be implemented as a PTP grandmaster.

(14) In this embodiment, the timing master **120** comprises one or more network interfaces **122** to communicatively couple the timing master **120** to the fronthaul **116**. In this particular embodiment, where the fronthaul **120** is implemented using a switched Ethernet network, the one or more network interfaces **122** comprise one or more Ethernet network interfaces.

(15) In the embodiment shown in FIG. **1**, the timing master **120** further comprises at least one programmable processor or other programmable device **124** coupled to the one or more network interfaces **122**. The programmable device **124** is configured to execute software **126**. The software **126** can be implemented as program instructions or configuration logic that are stored (or otherwise embodied) on an appropriate non-transitory storage medium or media **128** from which at least a portion of the program instructions or configuration logic are read by one or more programmable devices **124** for execution thereby or configuration thereof. The software **126** is configured to cause the timing master **120** to carry out at least some of the functions described here as being performed by it. Although the storage medium **128** is shown in FIG. **1** as being included in the timing master **120**, it is to be understood that remote storage media (for example, storage media that is accessible over a network) and/or removable media can also be used. The timing master **120** also comprises memory **130** for storing the program instructions or configuration logic and/or any related data while the functions implemented by the software **126** are performed.

(16) In the embodiment shown in FIG. **1**, the timing master **120** further comprises a global navigation satellite system (GNSS) receiver **132** coupled to at least one programmable device **124**. An antenna **133** is coupled to the GNSS receiver **132**. The GNSS receiver **132** is configured to receive and process signals broadcasted by satellites in order to determine the precise time and to synchronize the local clock **134** included in the timing master **120** to the precise time. The GNSS receiver **132** can be implemented so as to implement one of more of a Global Positioning System (GPS) GNSS receiver, a Global Navigation Satellite System (GLONASS) GNSS receiver, a Galileo GNSS receiver, and a BeiDou Navigation Satellite System (BDS) GNSS receiver. Other types of GNSS receivers **132** can be used.

(17) Each RU **108** is synchronized with its associated DU **106**. That is, the local clock in each RU **108** is synchronized with the local clock in the associated DU **106**. In this embodiment, this is done by synchronizing the DU **106** and each RU **108** to the timing master **120** using the Institute of Electrical and Electronics Engineers (IEEE) 1588 Precision Time Protocol (PTP). For example, the software **126** executed by the timing master **120** can implement a PTP grandmaster that is configured to communicate PTP Synch packets from the timing master **120** to at least some of the nodes of the RAN **100** (for example, the DU **106** and RU **108**) over the fronthaul network **116** at a Synch packet transmission rate (N_s), which is also referred to here as the “synchronization rate.” The synchronization rate N_s specifies the number of PTP Synch packets transmitted each second.

(18) As noted above, specifications promulgated by the O-RAN Alliance recommend the use of a Synch packet transmission rate N_s that is a power of 2. This may result in every Synch packet contending with an IQ packet transmitted from the DU **106** over the fronthaul **116**, which will impair the performance of the PTP synchronization process.

(19) In order to avoid this issue with using the recommended synchronization packet transmission rates, one or more of the DU **106** and/or the timing master **120** can be configured to avoid contention between IQ packets and at least some of the PTP Synch packets that are communicated

over the fronthaul **116**.

(20) FIG. 2 comprises a high-level flowchart illustrating one exemplary embodiment of a method **200** of operating a radio access network (RAN) in which contention between IQ packets and at least some synchronization packets communicated over the fronthaul **116** is avoided.

(21) The blocks of the flow diagram shown in FIG. 2 have been arranged in a generally sequential manner for ease of explanation; however, it is to be understood that this arrangement is merely exemplary, and it should be recognized that the processing associated with method **200** (and the blocks shown in FIG. 2) can occur in a different order (for example, where at least some of the processing associated with the blocks is performed in parallel and/or in an event-driven manner). Also, most standard exception handling is not described for ease of explanation; however, it is to be understood that method **200** can and typically would include such exception handling.

(22) The embodiment of method **200** shown in FIG. 2 is described here as being implemented in the RAN **100** of FIG. 1, though it is to be understood that other embodiments can be implemented in other ways.

(23) Method **200** comprises, while communicating IQ packets over the fronthaul network **116** to support wireless communication with user equipment **112** using a predetermined slot rate R_{slot} (block **202**), communicating synchronization packets from the timing master **120** to at least some of the nodes of the RAN **100** over the fronthaul network **116** with the respective transmission offsets modulo the slot duration of the synchronization packets transmitted during any given second being unique and distributed throughout the slot period (block **204**).

(24) In this example, the synchronization packets comprise PTP Synch packets that the timing master **120** communicates to the DU **106** and the RUs **108** over the switched Ethernet network used to implement the fronthaul network **116**.

(25) As noted above, the slot rate has an associate slot period specifying a duration of each slot (also referred to here as the “slot duration”). As used here, the “transmission offset” modulo the slot duration of a given synchronization packet transmitted by the timing master **120** is the difference between the time that synchronization packet is sent and the start of the slot during which that packet was sent.

(26) As noted above, with a slot rate of 8000 slots per second (corresponding to a 5G numerology of 3) and a synchronization packet transmission rate of 128 packets per second, the number of slots between consecutive synchronization packets would be 62.5 slots ($8000/128$) and synchronization packets would be transmitted at only two transmission offsets—either a transmission offset corresponding to the beginning of the slot or a transmission offset corresponding to the midpoint of the slot. If the transmission time (duration) for the burst of IQ packets for each slot is $T > 0.5$ slots and $0 \leq (\text{theta modulo } 1) \leq T - 0.5$, it is seen that all of the 128 synchronization packets will experience contention with IQ packets. However, if the synchronization packets are transmitted in a manner that the respective transmission offsets (modulo the slot duration) of the synchronization packets transmitted during any given second are unique and distributed throughout the slot period, there should be a number of synchronization packets that are transmitted during the portion of a slot when it is not likely that synchronization packets will experience contention with IQ packets. This will enable the synchronization process to be successfully performed.

(27) In one implementation of method **200**, the timing master **120** is configured to transmit the synchronization packets in a way so that the amount of time between the successive synchronization packets is constant and determined by the synchronization packet transmission rate. The timing master **120** is configured to use a synchronization packet transmission rate that results in the synchronization packets being transmitted from the timing master **120** with the respective transmission offsets of the synchronization packets transmitted during any given second being unique and uniformly distributed throughout the slot period. For example, for the set of synchronization packet transmission rates N_s from 1 to 128 (that is, for $N_s \in \{1, 2, 3, \dots, 127, 128\}$), there are 51 synchronization packet transmission rates N_s that will result in the

synchronization packets being transmitted from the timing master **120** with the respective transmission offsets of the synchronization packets transmitted during any given second being unique and uniformly distributed throughout the slot period. The 51 “best” synchronization packet transmission rates (in packets per second) are the following: 1, 3, 7, 9, 11, 13, 17, 19, 21, 23, 27, 29, 31, 33, 37, 39, 41, 43, 47, 49, 51, 53, 57, 59, 61, 63, 67, 69, 71, 73, 77, 79, 81, 83, 87, 89, 91, 93, 97, 99, 101, 103, 107, 109, 111, 113, 117, 119, 121, 123, and 127.

(28) In a second implementation, the timing master **120** is configured to randomize the transmission offsets of the synchronization packets uniformly over the slot period. For example, the timing master **120** can be configured so that the absolute transmission start time for the n th synchronization packet is randomized uniformly on the interval $n/N_s \pm 1/(2 R_{\text{slot}})$.

(29) FIG. 3 comprises a high-level flowchart illustrating one exemplary embodiment of a method **200** of operating a radio access network (RAN) in which contention between IQ packets and at least some synchronization packets communicated over the fronthaul **116** is avoided.

(30) The blocks of the flow diagram shown in FIG. 3 have been arranged in a generally sequential manner for ease of explanation; however, it is to be understood that this arrangement is merely exemplary, and it should be recognized that the processing associated with method **300** (and the blocks shown in FIG. 3) can occur in a different order (for example, where at least some of the processing associated with the blocks is performed in parallel and/or in an event-driven manner). Also, most standard exception handling is not described for ease of explanation; however, it is to be understood that method **300** can and typically would include such exception handling.

(31) Method **300** comprises, while receiving, by a node of a RAN **100** (such as the DU **1060**, synchronization packets transmitted from the timing master **120** over the fronthaul network **116** at a synchronization packet transmission rate and phase (block **302**), discovering, by that node, the synchronization packet transmission phase of the synchronization packets being transmitted (block **304**) and selectively delaying transmission of at least some in-phase and quadrature (IQ) data over the fronthaul network **116** in order to avoid contention with at least some of the synchronization packets transmitted from the timing master **120** over the fronthaul network **116** (block **306**).

(32) In one implementation, this is done by the DU **106**. The DU **106** is configured to discover the synchronization packet transmission phase of the synchronization packets being transmitted over the fronthaul **116** by tracking the times at which at least some of the synchronization packets are received and then determining the synchronization packet transmission phases from these tracked times (the synchronization packet transmission phase is the transmission time for synchronization packet modulo the slot duration). Then, the DU **106** uses the slot rate, the synchronization packet transmission rate, and phase to determine the transmission time offset for each synchronization packet transmitted during a given second and uses these transmission time offsets to determine if at least some of the synchronization packets transmitted from the timing master **120** over the fronthaul network **116** will experience contention with IQ packets transmitted from the DU **106**. If that is the case, the DU **106** delays the transmission of at least some IQ data over the fronthaul network **116** in order to create gaps in the IQ bursts and avoid contention with at least some of the synchronization packets transmitted from the timing master **120**.

(33) Performing the processing associated with method **300** ensures that at least some of the synchronization packets transmitted from the timing master **120** over the fronthaul network **116** will not experience contention with IQ packets transmitted from the DU **106**, which enables the synchronization process to be successfully performed.

(34) Other embodiments can be implemented in other ways.

(35) A number of embodiments of the invention defined by the following claims have been described. Nevertheless, it will be understood that various modifications to the described embodiments may be made without departing from the spirit and scope of the claimed invention. Accordingly, other embodiments are within the scope of the following claims.

Example Embodiments

(36) Example 1 includes a timing master for use in a radio access network (RAN) comprising a plurality of nodes, at least some of which are communicatively coupled to each other via a fronthaul network, wherein the RAN is configured to wirelessly communicate with user equipment (UE) using a predetermined slot rate having an associated slot period specifying a slot duration of each slot, the timing master comprising: one or more network interfaces to communicatively couple the timing master to the fronthaul network; at least one programmable device coupled to the one or more network interfaces; a global navigation satellite system (GNSS) receiver coupled to the at least one programmable processor; and a local clock coupled to the at least one programmable device; wherein the timing master is configured to communicate synchronization packets from the timing master to at least some of the nodes of the RAN over the fronthaul network at a synchronization packet transmission rate, the synchronization packet transmission rate specifying a number of synchronization packets transmitted each second, wherein each of said number of synchronization packets transmitted each second is transmitted at a respective transmission time offset relative to the slot period; and wherein the timing master is configured so that the respective transmission time offsets modulo the slot duration for said number of synchronization packets transmitted during each second are unique and distributed throughout the slot period.

(37) Example 2 includes the timing master of Example 1, wherein the one or more network interfaces comprises one or more Ethernet network interfaces to communicatively couple the timing master to a switched Ethernet fronthaul network.

(38) Example 3 includes the timing master of any of Examples 1-2, wherein the timing master is configured to implement an Institute of Electrical and Electronics Engineers (IEEE) 1588 Precision Time Protocol (PTP), wherein the synchronization packets comprise PTP synchronization packets.

(39) Example 4 includes the timing master of Example 3, wherein the timing master comprises a PTP grandmaster.

(40) Example 5 includes the timing master of any of Examples 1-4, wherein the GNSS receiver comprises at least one of a Global Positioning System (GPS) GNSS receiver, a Global Navigation Satellite System (GLONASS) GNSS receiver, a Galileo GNSS receiver, and a BeiDou Navigation Satellite System (BDS) GNSS receiver.

(41) Example 6 includes the timing master of any of Examples 1-5, wherein the timing master is configured to transmit the synchronization packets so that the amount of time between the successive synchronization packets is constant and determined by the synchronization packet transmission rate, wherein the synchronization packet transmission rate that is used results in the synchronization packets being transmitted from the timing master with the respective transmission offsets modulo slot duration of the synchronization packets transmitted during any given second being unique and uniformly distributed throughout the slot period.

(42) Example 7 includes the timing master of Example 6, wherein the synchronization packet transmission rate comprises one of: 1 packet per second, 3 packets per second, 7 packets per second, 9 packets per second, 11 packets per second, 13 packets per second, 17 packets per second, 19 packets per second, 21 packets per second, 23 packets per second, 27 packets per second, 29 packets per second, 31 packets per second, 33 packets per second, 37 packets per second, 39 packets per second, 41 packets per second, 43 packets per second, 47 packets per second, 49 packets per second, 51 packets per second, 53 packets per second, 57 packets per second, 59 packets per second, 61 packets per second, 63 packets per second, 67 packets per second, 69 packets per second, 71 packets per second, 73 packets per second, 77 packets per second, 79 packets per second, 81 packets per second, 83 packets per second, 87 packets per second, 89 packets per second, 91 packets per second, 93 packets per second, 97 packets per second, 99 packets per second, 101 packets per second, 103 packets per second, 107 packets per second, 109 packets per second, 111 packets per second, 113 packets per second, 117 packets per second, 119 packets per second, 121 packets per second, 123 packets per second, and 127 packets per second.

(43) Example 8 includes the timing master of any of Examples 1-7, wherein the timing master is

configured to randomize the transmission offsets of the synchronization packets uniformly over the slot period.

(44) Example 9 includes the timing master of Example 8, wherein the timing master is configured to randomize the transmission time of the synchronization packets at the synchronization packet transmission rate N_s uniformly over the slot period by randomizing the transmission time of the n th synchronization packet for each second uniformly over a period that is n/N_s plus or minus one-half of the slot duration of the slot period.

(45) Example 10 includes a system comprising: a radio access network (RAN) comprising a plurality of nodes, at least some of which are communicatively coupled to each other via a fronthaul network; and a timing master communicatively coupled to the nodes of the RAN via the fronthaul network; wherein the RAN is configured to wirelessly communicate with user equipment (UE) using a predetermined slot rate having an associated slot period specifying a slot duration of each slot; wherein the system is configured to communicate synchronization packets from the timing master to at least some of the nodes of the RAN over the fronthaul network at a synchronization packet transmission rate, the synchronization packet transmission rate specifying a number of synchronization packets transmitted each second, wherein each of said number of synchronization packets transmitted each second is transmitted at a respective transmission time offset relative to the slot period; and wherein the system is configured so that the respective transmission time offsets modulo the slot duration for said number of synchronization packets transmitted during each second are unique and distributed throughout the slot period.

(46) Example 11 includes the system of Example 10, wherein the RAN comprises: a centralized unit (CU); a distributed unit (DU) communicatively coupled to the CU; and a plurality of remote units (RUs) communicatively coupled to the DU via the fronthaul network.

(47) Example 12 includes the system of any of Examples 10-11, wherein the fronthaul network comprises a switched Ethernet fronthaul network.

(48) Example 13 includes the system of any of Examples 10-12, wherein the synchronization packets comprise Institute of Electrical and Electronics Engineers (IEEE) 1588 Precision Time Protocol (PTP) synchronization packets.

(49) Example 14 includes the system of Example 13, wherein the timing master comprises a PTP grandmaster.

(50) Example 15 includes the system of any of Examples 10-14, wherein the system is configured to wirelessly communicate with the user equipment using a Fifth Generation (5G) wireless protocol.

(51) Example 16 includes the system of any of Examples 10-15, wherein the timing master is configured to transmit the synchronization packets so that the amount of time between the successive synchronization packets is constant and determined by the synchronization packet transmission rate, wherein the synchronization packet transmission rate that is used results in the synchronization packets being transmitted from the timing master with the respective transmission offsets modulo the slot duration of the synchronization packets transmitted during any given second being unique and uniformly distributed throughout the slot period.

(52) Example 17 includes the system of Example 16, wherein the synchronization packet transmission rate comprises one of: 1 packet per second, 3 packets per second, 7 packets per second, 9 packets per second, 11 packets per second, 13 packets per second, 17 packets per second, 19 packets per second, 21 packets per second, 23 packets per second, 27 packets per second, 29 packets per second, 31 packets per second, 33 packets per second, 37 packets per second, 39 packets per second, 41 packets per second, 43 packets per second, 47 packets per second, 49 packets per second, 51 packets per second, 53 packets per second, 57 packets per second, 59 packets per second, 61 packets per second, 63 packets per second, 67 packets per second, 69 packets per second, 71 packets per second, 73 packets per second, 77 packets per second, 79 packets per second, 81 packets per second, 83 packets per second, 87 packets per second, 89

packets per second, 91 packets per second, 93 packets per second, 97 packets per second, 99 packets per second, 101 packets per second, 103 packets per second, 107 packets per second, 109 packets per second, 111 packets per second, 113 packets per second, 117 packets per second, 119 packets per second, 121 packets per second, 123 packets per second, and 127 packets per second. (53) Example 18 includes the system of any of Examples 10-17, wherein the timing master is configured to randomize the transmission offsets of the synchronization packets uniformly over the slot period.

(54) Example 19 includes the system of Example 18, wherein the timing master is configured to randomize the transmission time of the synchronization packets at the synchronization packet transmission rate N_s uniformly over the slot period by randomizing the transmission time of the n th synchronization packet for each second uniformly over a period that is n/N_s plus or minus one-half of the slot duration of the slot period.

(55) Example 20 includes a method of operating a radio access network (RAN) comprising a plurality of nodes, at least some of which are communicatively coupled to each other via a fronthaul network, the method comprising: wirelessly communicating with user equipment (UE) using a predetermined slot rate having an associated slot period specifying a slot duration of each slot; and communicating synchronization packets from the timing master to at least some of the nodes of the RAN over the fronthaul network at a synchronization packet transmission rate, the synchronization packet transmission rate specifying a number of synchronization packets transmitted each second, wherein each of said number of synchronization packets transmitted each second is transmitted at a respective transmission time offset relative to the slot period; and wherein the respective transmission time offsets modulo the slot duration for said number of synchronization packets transmitted during each second are unique and distributed throughout the slot period.

(56) Example 21 includes the method of Example 20, wherein communicating the synchronization packets from the timing master to at least some of the nodes of the RAN over the fronthaul network at the synchronization packet transmission rate comprises using a synchronization packet transmission rate that results in the synchronization packets being transmitted from the timing master so that the amount of time between the successive synchronization packets is constant and determined by the synchronization packet transmission rate, wherein the synchronization packet transmission rate that is used results in the synchronization packets being transmitted from the timing master with the respective transmission offsets of the synchronization packets transmitted during any given second being unique and uniformly distributed throughout the slot period.

(57) Example 22 includes the method of Example 21, wherein the synchronization packet transmission rate comprises one of: 1 packet per second, 3 packets per second, 7 packets per second, 9 packets per second, 11 packets per second, 13 packets per second, 17 packets per second, 19 packets per second, 21 packets per second, 23 packets per second, 27 packets per second, 29 packets per second, 31 packets per second, 33 packets per second, 37 packets per second, 39 packets per second, 41 packets per second, 43 packets per second, 47 packets per second, 49 packets per second, 51 packets per second, 53 packets per second, 57 packets per second, 59 packets per second, 61 packets per second, 63 packets per second, 67 packets per second, 69 packets per second, 71 packets per second, 73 packets per second, 77 packets per second, 79 packets per second, 81 packets per second, 83 packets per second, 87 packets per second, 89 packets per second, 91 packets per second, 93 packets per second, 97 packets per second, 99 packets per second, 101 packets per second, 103 packets per second, 107 packets per second, 109 packets per second, 111 packets per second, 113 packets per second, 117 packets per second, 119 packets per second, 121 packets per second, 123 packets per second, and 127 packets per second.

(58) Example 23 includes the method of any of Examples 20-22, wherein communicating the synchronization packets from the timing master to at least some of the nodes of the RAN over the fronthaul network at the synchronization packet transmission rate comprises randomizing the

transmission offsets of the synchronization packets uniformly over the slot period.

(59) Example 24 includes the method of any of Examples 20-23, wherein the timing master is configured to randomize the transmission time of the synchronization packets at the synchronization packet transmission rate N_s uniformly over the slot period by randomizing the transmission time of the n th synchronization packet for each second uniformly over a period that is n/N_s plus or minus one-half of the slot duration of the slot period.

(60) Example 25 includes a radio access network (RAN) system comprising: a plurality of nodes, at least some of which are communicatively coupled to each other via a fronthaul network; wherein the RAN system is configured to wirelessly communicate with user equipment (UE); and wherein the system is configured so that, while at least one of the nodes of the RAN system receives synchronization packets transmitted from a timing master over the fronthaul network at a synchronization packet transmission rate and synchronization packet phase, said at least one of the nodes discovers the synchronization packet transmission phase of the synchronization packets being transmitted from the timing master and selectively delays transmission of at least some in-phase and quadrature (IQ) data over the fronthaul network in order to avoid contention with at least some of the synchronization packets transmitted from the timing master over the fronthaul network.

(61) Example 26 includes the RAN system of Example 25, wherein the plurality of nodes comprises: a distributed unit (DU); and a plurality of remote units (RUs) communicatively coupled to the DU via the fronthaul network.

(62) Example 27 includes the RAN system of Example 26, wherein said at least one of the nodes comprises the DU.

(63) Example 28 includes the RAN system of any of Examples 26-27, wherein the plurality of nodes further comprises a centralized unit (CU) that is communicatively coupled to the DU.

(64) Example 29 includes the RAN system of any of Examples 25-28, wherein said at least one of the nodes discovers the synchronization packet transmission phase of the synchronization packets being transmitted from the timing master by: tracking the times at which at least some of the synchronization packets are received; and determining the synchronization packet transmission phase from the tracked times.

(65) Example 30 includes the RAN system of Example 29, wherein said at least one of the nodes selectively delays transmission of at least some IQ data over the fronthaul network in order to avoid contention with at least some of the synchronization packets transmitted from the timing master over the fronthaul network by: using the slot rate, the synchronization packet transmission rate, and the discovered synchronization packet transmission phase to determine the transmission time offset for each synchronization packet transmitted during a given second; using the transmission time offsets to determine if at least some of the synchronization packets transmitted from the timing master over the fronthaul network will experience contention with IQ packets transmitted from said at least one of the nodes over the fronthaul network; and in response to determining that at least some of the synchronization packets transmitted from the timing master over the fronthaul network will experience contention with IQ packets transmitted from said at least one of the nodes over the fronthaul network, delaying transmission of at least some IQ data over the fronthaul network in order to avoid contention with at least some of the synchronization packets transmitted from the timing master.

(66) Example 31 includes the RAN system of any of Examples 25-30, wherein the fronthaul network comprises a switched Ethernet fronthaul network.

(67) Example 32 includes the RAN system of any of Examples 25-31, wherein the synchronization packets comprise Institute of Electrical and Electronics Engineers (IEEE) 1588 Precision Time Protocol (PTP) synchronization packets.

(68) Example 33 includes the RAN system of Example 32, wherein the timing master comprises a PTP grandmaster.

(69) Example 34 includes the RAN system of any of Examples 25-33, wherein the RAN system is

configured to wirelessly communicate with the user equipment using a Fifth Generation (5G) wireless protocol.

(70) Example 35 includes a method of operating a radio access network (RAN) comprising a plurality of nodes, at least some of which are communicatively coupled to each other via a fronthaul network, wherein the RAN is configured to wirelessly communicate with user equipment (UE), the method comprises: while at least one of the nodes of the RAN system receives synchronization packets transmitted from a timing master over the fronthaul network at a synchronization packet transmission rate and a synchronization packet transmission phase, discovering, by said at least one of the nodes, the synchronization packet transmission phase of the synchronization packets being transmitted from the timing master; and selectively delaying, by said at least one of the nodes, transmission of at least some in-phase and quadrature (IQ) data over the fronthaul network in order to avoid contention with at least some of the synchronization packets transmitted from the timing master over the fronthaul network.

(71) Example 36 includes the method of Examples 35, wherein discovering, by said at least one of the nodes, the synchronization packet transmission phase of the synchronization packets being transmitted from the timing master comprises: tracking the times at which at least some of the synchronization packets are received; and determining the synchronization packet transmission phase from the tracked times.

(72) Example 37 includes the method of Example 36, wherein selectively delaying, by said at least one of the nodes, transmission of at least some IQ data over the fronthaul network in order to avoid contention with at least some of the synchronization packets transmitted from the timing master over the fronthaul network comprises: using the slot rate, the synchronization packet transmission rate, and the discovered synchronization packet transmission phase to determine the transmission time offset for each synchronization packet transmitted during a given second; using the transmission time offsets to determine if at least some of the synchronization packets transmitted from the timing master over the fronthaul network will experience contention with IQ packets transmitted from said at least one of the nodes over the fronthaul network; and in response to determining that at least some of the synchronization packets transmitted from the timing master over the fronthaul network will experience contention with IQ packets transmitted from said at least one of the nodes over the fronthaul network, delaying transmission of at least some IQ data over the fronthaul network in order to avoid contention with at least some of the synchronization packets transmitted from the timing master.

(73) Example 38 includes the method of any of Examples 35-37, wherein the fronthaul network comprises a switched Ethernet fronthaul network.

(74) Example 39 includes the method of any of Examples 35-38, wherein the synchronization packets comprise Institute of Electrical and Electronics Engineers (IEEE) 1588 Precision Time Protocol (PTP) synchronization packets.

(75) Example 40 includes the method of Example 39, wherein the timing master comprises a PTP grandmaster.

(76) Example 41 includes the method of any of Examples 35-40, wherein the RAN is configured to wirelessly communicate with the user equipment using a Fifth Generation (5G) wireless protocol.

Claims

1. A radio access network (RAN) system comprising: a plurality of nodes, at least some of which are communicatively coupled to each other via a fronthaul network; wherein the RAN system is configured to wirelessly communicate with user equipment (UE); and wherein the system is configured so that, while at least one of the nodes of the RAN system receives synchronization packets transmitted from a timing master over the fronthaul network at a synchronization packet transmission rate and synchronization packet phase, said at least one of the nodes discovers the

- synchronization packet transmission phase of the synchronization packets being transmitted from the timing master and selectively delays, using the discovered synchronization packet transmission phase, transmission of at least some in-phase and quadrature (IQ) data over the fronthaul network in order to avoid contention with at least some of the synchronization packets transmitted from the timing master over the fronthaul network.
2. The RAN system of claim 1, wherein the plurality of nodes comprises: a distributed unit (DU); and a plurality of remote units (RUs) communicatively coupled to the DU via the fronthaul network.
 3. The RAN system of claim 2, wherein said at least one of the nodes comprises the DU.
 4. The RAN system of claim 2, wherein the plurality of nodes further comprises a centralized unit (CU) that is communicatively coupled to the DU.
 5. The RAN system of claim 1, wherein said at least one of the nodes discovers the synchronization packet transmission phase of the synchronization packets being transmitted from the timing master by: tracking the times at which at least some of the synchronization packets are received; and determining the synchronization packet transmission phase from the tracked times.
 6. The RAN system of claim 5, wherein said at least one of the nodes selectively delays, using the discovered synchronization packet transmission phase, transmission of at least some IQ data over the fronthaul network in order to avoid contention with at least some of the synchronization packets transmitted from the timing master over the fronthaul network by: using the slot rate, the synchronization packet transmission rate, and the discovered synchronization packet transmission phase to determine the transmission time offset for each synchronization packet transmitted during a given second; using the transmission time offsets to determine if at least some of the synchronization packets transmitted from the timing master over the fronthaul network will experience contention with IQ packets transmitted from said at least one of the nodes over the fronthaul network; and in response to determining that at least some of the synchronization packets transmitted from the timing master over the fronthaul network will experience contention with IQ packets transmitted from said at least one of the nodes over the fronthaul network, delaying transmission of at least some IQ data over the fronthaul network in order to avoid contention with at least some of the synchronization packets transmitted from the timing master.
 7. The RAN system of claim 1, wherein the fronthaul network comprises a switched Ethernet fronthaul network.
 8. The RAN system of claim 1, wherein the synchronization packets comprise Institute of Electrical and Electronics Engineers (IEEE) 1588 Precision Time Protocol (PTP) synchronization packets.
 9. The RAN system of claim 8, wherein the timing master comprises a PTP grandmaster.
 10. The RAN system of claim 1, wherein the RAN system is configured to wirelessly communicate with the user equipment using a Fifth Generation (5G) wireless protocol.
 11. A method of operating a radio access network (RAN) comprising a plurality of nodes, at least some of which are communicatively coupled to each other via a fronthaul network, wherein the RAN is configured to wirelessly communicate with user equipment (UE), the method comprises: while at least one of the nodes of the RAN system receives synchronization packets transmitted from a timing master over the fronthaul network at a synchronization packet transmission rate and a synchronization packet transmission phase, discovering, by said at least one of the nodes, the synchronization packet transmission phase of the synchronization packets being transmitted from the timing master; and selectively delaying, by said at least one of the nodes using the discovered synchronization packet transmission phase, transmission of at least some in-phase and quadrature (IQ) data over the fronthaul network in order to avoid contention with at least some of the synchronization packets transmitted from the timing master over the fronthaul network.
 12. The method of claim 11, wherein discovering, by said at least one of the nodes, the synchronization packet transmission phase of the synchronization packets being transmitted from the timing master comprises: tracking the times at which at least some of the synchronization

packets are received; and determining the synchronization packet transmission phase from the tracked times.

13. The method of claim 12, wherein selectively delaying, by said at least one of the nodes using the discovered synchronization packet transmission phase, transmission of at least some IQ data over the fronthaul network in order to avoid contention with at least some of the synchronization packets transmitted from the timing master over the fronthaul network comprises: using the slot rate, the synchronization packet transmission rate, and the discovered synchronization packet transmission phase to determine the transmission time offset for each synchronization packet transmitted during a given second; using the transmission time offsets to determine if at least some of the synchronization packets transmitted from the timing master over the fronthaul network will experience contention with IQ packets transmitted from said at least one of the nodes over the fronthaul network; and in response to determining that at least some of the synchronization packets transmitted from the timing master over the fronthaul network will experience contention with IQ packets transmitted from said at least one of the nodes over the fronthaul network, delaying transmission of at least some IQ data over the fronthaul network in order to avoid contention with at least some of the synchronization packets transmitted from the timing master.

14. The method of claim 11, wherein the fronthaul network comprises a switched Ethernet fronthaul network.

15. The method of claim 11, wherein the synchronization packets comprise Institute of Electrical and Electronics Engineers (IEEE) 1588 Precision Time Protocol (PTP) synchronization packets.

16. The method of claim 15, wherein the timing master comprises a PTP grandmaster.

17. The method of claim 11, wherein the RAN is configured to wirelessly communicate with the user equipment using a Fifth Generation (5G) wireless protocol.
